

Two-Digit Subtraction with Place-Value Models

Answer Key

Grades 2–3

Quick Answers

1. 24	3. 30	5. 37	7. 16
2. 33	4. 25	6. 40, 38	8. 34

Curriculum

Level: Grades 2–3

2.NBT.B.5 – *problems 1, 2, 4, 5, 8*

3.NBT.A.2 – *problems 3, 6, 7*

Difficulty 1–4 of 5 across 9 tagged checks.

Common wrong answers

4. *If they got 35: took the smaller ones digit away from the larger one instead of regrouping a ten*
5. *If they got 43: took the smaller ones digit away from the larger one instead of regrouping a ten*
8. *If they got 46: took the smaller ones digit away from the larger one in each column instead of regrouping*
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1. Take away with blocks: $47 - 23$. The picture shows 4 ten rods and 7 ones cubes. Crossing out 23 means crossing out 2 ten rods and 3 ones cubes. Tens left: $4 - 2 = 2$ rods, which is 20. Ones left: $7 - 3 = 4$ cubes. Put the parts back together: $20 + 4$. There were enough ones to take away, so no ten had to be broken. The difference is 24.

2. Take away with blocks: $68 - 35$. The picture shows 6 ten rods and 8 ones cubes. Crossing out 35 removes 3 ten rods and 5 ones cubes. Tens left: $6 - 3 = 3$ rods, which is 30. Ones left: $8 - 5 = 3$ cubes. Together that is $30 + 3$. Again there were enough ones, so nothing had to be broken. The difference is 33.

3. Round to the nearest ten: estimate $61 - 28$. 61 is between 60 and 70, and it is only 1 past 60, so it rounds down to 60. 28 is between 20 and 30, and it is 8 past 20, so it rounds up to 30. Now subtract the rounded numbers: $60 - 30 = 30$. So the answer to $61 - 28$ should be about 30 — a student who wrote 89 or 3 would know at once that something went wrong. The estimate is 30.

4. Break a ten: $52 - 27$. Expanded form: $52 = 50 + 2$ and $27 = 20 + 7$. There are only 2 ones and 7 ones must go, so break one ten rod into ten ones: $52 = 40 + 12$. Now take the tens away: $40 - 20 = 20$. Then take the ones away: $12 - 7 = 5$. Put the parts back together: $20 + 5$. Check by adding back: $27 + 25 = 52$. The difference is 25.

5. Break a ten: $83 - 46$. Expanded form: $83 = 80 + 3$ and $46 = 40 + 6$. There are only 3 ones and 6 ones must go, so break one ten rod: $83 = 70 + 13$. Tens: $70 - 40 = 30$. Ones: $13 - 6 = 7$. Together: $30 + 7$. Check by adding back: $46 + 37 = 83$. The difference is 37.

6. Estimate, then check: $85 - 47$. Estimate first. 85 rounds up to 90 and 47 rounds up to 50, so the estimate is $90 - 50 = 40$, which is 40.

Now the exact answer. Expanded form: $85 = 80 + 5$ and $47 = 40 + 7$. There are not enough ones, so break a ten: $85 = 70 + 15$. Tens: $70 - 40 = 30$. Ones: $15 - 7 = 8$. Together: $30 + 8$. Check by adding back: $47 + 38 = 85$. The exact difference is 38.

The exact answer is only 2 away from the estimate, so the work is reasonable.

7. Sticker album: Nadia had 74 stickers and glued 58 of them down, so the question asks how many are left — that is $74 - 58$. Draw 7 ten rods and 4 ones cubes. Only 4 ones are showing and 8 ones must go, so break one ten rod into ten ones: $74 = 60 + 14$. Tens: $60 - 50 = 10$. Ones: $14 - 8 = 6$. Together: $10 + 6$. Check by adding back: $58 + 16 = 74$. Nadia has 16 stickers left.

8. Find the mistake: $63 - 29$. Theo wrote $9 - 3 = 6$ for the ones column. He turned the subtraction around because 3 was smaller than 9; the ones digit of 63 is 3, and $3 - 9$ cannot be done without regrouping. That is the whole error, and it made his answer too big. The correct move is to break one of the 6 tens: $63 = 50 + 13$. Tens: $50 - 20 = 30$. Ones: $13 - 9 = 4$. Together: $30 + 4$. Check by adding back: $29 + 34 = 63$. The correct difference is 34, not 46.